

University of Dhaka
Affiliated Engineering Colleges
Department of Computer Science and Engineering
1st Year 2nd Semester B.Sc. Examination, 2024
PHY – 1203, Physics

Total Marks: 70

Time: 3 Hours

(Answer any 5 (Five) of the following Questions)

1. a) Define plane progressive wave and torsional pendulum. 2+2
 b) Derive the differential equation of a plane progressive wave. 5
 c) For adiabatic process prove that, $TV^{\gamma-1} = \text{constant}$. Define c_p and c_v . 5

2. a) What are mechanical and non-mechanical waves. Give their examples. 2
 b) Establish the differential equation of a damped harmonic oscillator and solve it. 5
 c) Distinguish between transverse wave and longitudinal wave. 2
 d) The displacement of an oscillating particle at an instant t is given by $y = a \cos \omega t + b \sin \omega t$ show that it is executing a simple harmonic motion. If $a = 5 \text{ cm}$ and $b = 12 \text{ cm}$ and $\omega = 4 \text{ radian/see}$. Calculate (i) the amplitude (ii) the time period (iii) the maximum velocity and (iv) the maximum acceleration of the particle. 5

3. a) Define (i) Coherent source (ii) Slit (iii) Huygens principle (iv) Double reflection 4
 b) What is interference? Find the condition for constructive interference and Destructive interference 5
 c) In a Young's double slit experiment, the separation between the sources is 0.18 mm . The fringes are observed on a screen 90 cm away. If with certain monochromatic source of light, the third bright fringe is situated at a distance of 8.1 mm from the central bright fringe, find the wavelength of light. 5

4. a) What are the differences between polarized light and unpolarized light? 4
 b) From Newton's rings prove that the radius of the n^{th} dark ring is proportional to the root of wavelength. 6
 c) Define refractive index. In the experiment of Newton's rings, diameter of the eighth ring is 0.4 cm and of the third ring is 0.2 cm . Radius of curvature of the lens is 101 cm . Calculate the wavelength of the light used. 4

5. a) Define: i) Joule's equivalent ii) thermodynamic function iii) thermodynamic system iv) internal energy. 4
 b) What is entropy? Show that entropy increases in an irreversible process whereas remains constant in a reversible process. 6
 c) A motor tyre is pumped to a pressure of 2 atmospheres at 15°C , when it suddenly bursts. Calculate the resulting drop in temperature, given $\gamma = 1.4$. 4

6. a) Make a difference between semiconductor and insulator on the view of band theory. 4
 b) What is Miller index? Show that for cubic crystal structure, the interplanter distance is
$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$
 4
 c) Sodium is a bcc crystal. Its density is $9.6 \times 10^2 \text{ kg/m}^3$ and atomic mass is 23. Calculate the lattice constant for a sodium crystal. 4
 d) Write down Carnot's theorem. 2

7. a) Establish the relation between particle velocity and wave velocity 4
 b) Describe the Brewster's law. From Brewster's law show that the polarized angle of incidence between the reflected and refracted rays are 90° apart. 5
 c) Show that for fcc crystal structure the lattice constant is given by $a_{\text{fcc}} = \frac{4r}{\sqrt{2}}$. 5

$\delta T = \frac{2T}{\gamma - 1}$

University of Dhaka
Affiliated Engineering Colleges
Department of Computer Science and Engineering
1st Year 2nd Semester B.Sc. Examination-2024
ENG-1205, Developing English Language Skills

Time: 3 Hours

Total Marks: 70

Part A (Comprehension: 15 marks)

1. Read the following passage carefully and answer the questions given below:

Man's intellect has a natural pride in its own aristocracy, which is the pride of its culture. Culture only acknowledges the excellence whose criticism is in its inner perfection, not in any external success. When this pride succumbs to some compulsion of necessity or lure of material advantage, it brings humiliation to the intellectual man. Modern India, through her very education, has been made to suffer this humiliation. Once she herself provided her children with a culture which was the product of her own ages of thought and creation. But it has been thrust aside, and we are made to tread the mill of passing examinations, not for learning anything, but for notifying that we are qualified for employments under organization conducted in English. "Our educated community is not a cultured community, but a community of qualified candidates". (Rabindranath Tagore)

Meanwhile the proportion of possible employments to the number of claimants has gradually been growing narrower, and the consequent disaffection has been widespread. At last the very authorities who are responsible for this are blaming their victims. Such is the perversity of human nature. It bears its worst grudge against those it has injured.....

In the Bengali language there is a modern maxim which can be translated, 'He who learns to read and write rides in a carriage and pair.' In English there is a similar proverb, 'Knowledge is power.' It is an offer of a prospective bribe to the student, a promise of an ulterior reward which is more important than knowledge itself.

Unfortunately, our very education has been successful in depriving us of our real initiative and our courage of thought. The training we get in our schools has the constant implication in it that it is not for us to produce but to borrow. And we are casting about to borrow our educational plans from European institutions. The trampled plants of Indian corn are dreaming of recouping their harvest from the neighbouring wheat fields. To change the figure, we forget that, for proficiency in walking, it is better to train the muscles of our own legs than to strut upon wooden ones of foreign make, although they clatter and cause more surprise at our skill in using them than if they were living and real.

a. Choose the correct answer:

05

i) Man's intellect has a---

- a) benevolence ☒ b) cruelty c) vanity ☒ d) dishonesty

ii) In the text 'The trampled plants of Indian corn' means-

- a) the labourers of India b) the agriculture of India
c) the students of India d) the economy of India

iii) The word 'humiliation' means-

- a) honour b) failing c) fun d) disgrace

iv) 'Maxim' means-

- a) idiom b) proverb c) phrase d) sentence

v) Man's intellectual has a natural pride in its "own" aristocracy, The underlined word is-

- a) noun b) adverb c) adjective d) pronoun

b. Answer the following questions:

05

- i) Why did Tagore call our educated community a community of qualified candidates?
- ii) What do you mean by- "Man's intellect has a natural pride in its own aristocracy"?
- iii) What does 'knowledge is power' mean?
- iv) Why is the disaffection becoming widespread?
- v) What does bring humiliation to intellectual man?

c. Write a summary of the passage.

05

Part B (Grammar: 25 marks)

2. Complete the text with suitable prepositions. 05
 A good student never learns things (a) ----- rote. He always adheres (b) ----- his studies. He does not hunt (c) ----- traditional guidebooks. He abides (d) ----- his teacher's advice and jots (e) ----- their lectures. He does not mix (f) ----- bad boys and while (g) ----- his time. He also behaves well (h) ----- the superiors and never looks down (i) ----- anybody. Everybody is pleased (j) ----- his conduct.
3. Fill in the blanks in the following text with articles(a/an/the) as necessary: 05
 Every student wants to do well in (a) ----- examination. But it is not (b) ----- easy work. (c) ----- student has to do something for this. From (d) ----- very beginning, he must be serious. He should read texts again and again. He must not memorize (e) ----- answer without knowing the meaning. He must not make notes from (f) ----- common source. He should have (g) ----- good (h) ----- command of English. By doing all these things (i) ----- student can hope to make (j) ----- excellent result.
4. Transform any five (5) out of the following sentences as directed: 05
 a) Cricket is very exciting game.(Exclamatory)
 b) We work hard to attain success in our Life. (Complex)
 c) The padma is the biggest of all the rivers in Bangladesh. (Positive)
 d) Nowadays books are published in a huge quantity. (Active)
 e) A house is the place where we live in. (Compound)
 f) He delivered a speech and everybody admired it. (Simple)
 g) If you don't eat good foods, you can't be healthy (compound)
5. Fill in the blanks with appropriate Phrases/words: 05
 a) Rupa _____ take up the profession of teaching than be a nurse.
 b) Edmund Burke was an orator, writer, and shrewd politician. He _____ in 1729 in England.
 c) _____ is difficult to regain lost health. So we should always take care of our health.
 d) _____ flows a river beside our village, the river is useful to us in many ways.
 e) Sarika is an examinee. She _____ study a lot to make a good result.
 f) _____ honesty _____? I think it to be a noble virtue.
 g) Rita is ill. We _____ go to see her. She will be happy to see us.
 h) The girls feel sick _____ they had food, the food was contaminated.
 i) _____ an elephant _____? I have never seen it before?
 j) He cannot teach in a high school, _____ in a university.
6. Correct the following sentences: 05
 a) He got himself married there.
 b) He can talk English fluently.
 c) The beggar asked for an alms.
 d) It is high time we work for the country.
 e) Neither of these two women are to be trusted.

Part C (Writing: 30 marks)

7. Write a Paragraph about "Problems of cyber security in Bangladesh". 08
8. Write a paragraph on "Causes and Effects of Climate Change" or "E-Learning". 07
9. Write an essay on "The necessity of Compulsory ICT Education in Bangladesh." 08
10. Write a report about "Food Adulteration and it's damning consequences". 07

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Department of Computer Science & Engineering
1st Year 2nd Semester, B.Sc. Examination-2024
MATH- 1204, Methods of Integration & Differential Calculus and Series

Total Marks: 70

Time: 3 Hours

(Answer any 5 (Five) of the following questions)

1. a) Define order & degree of ODE. Find the differential equation whose solution $v = \frac{A}{r} + B$ is where A & B are arbitrary constant. 5
- b) Find the ODE of $y = e^x(A\cos x + B\sin x)$ and hence the order & degree. 4
- c) Solve $D^2y + 4y = \sin 2x$. 5
2. a) State Simpson's one-third rule. Approximate $\int_0^2 \sqrt{x(4-x)} dx$ using the Simpson's one-third rule with $h = 0.2$. 4
- b) Calculate the entire arc length of the cardioid $r = a(1 + \cos \theta)$ and demonstrate that the arc of the upper half is bisected by $\theta = \frac{\pi}{3}$. 5
- c) Determine the surface area of the solid generated by revolving the parabola $y^2 = 4ax$ between $x = 0$ and $x = a$ about x -axis. 5
3. a) Solve $\cos^2 x \frac{dy}{dx} + y = \tan x$. 4
- b) Find the general solution & singular solution of $p = px + p - p^2$. 7
- c) Solve the given homogeneous linear differential equation $(D^2 + 6D + 25)y = 0$. 3
4. a) State and prove Bernoulli's equation. 4
- b) Show that the differential equation of the circle tangent to the x -axis at the origin is $(x^2 - y^2)dy - 2xydx = 0$. 5
- c) According to the U.S. Census Bureau, the world population in 2011 was 6.9 billion and growing at a rate of about 1.10% per year. Assuming an exponential growth model, estimate the world population at the beginning of the year 2030. 5
5. a) Integrate: i) $\int_{-\infty}^{\infty} \frac{x dx}{1+x^2}$; ii) $\int \frac{7x+4}{x^3-4x} dx$. 4+4
- b) A 30 volts electromotive force is applied to an LR series circuit in which the inductance is 0.2 henry and resistance is 50 ohms. Find the current $i(t)$ if $i(0)=0$. Determine the current after a long time. 6
6. a) Write Taylor's formula with remainder and expand $\ln x$ as a power series in $(x-2)$. 4
- b) Determine the n th degree Maclaurin polynomial of $\sin x$. 5
- c) State D'Alembert's ratio test and examine the convergence of the given series 5

$$x + \frac{1}{2} \times \frac{x^3}{3} + \frac{1.3}{2.4} \times \frac{x^5}{5} + \frac{1.3.5}{2.4.6} \times \frac{x^7}{7} + \dots \dots \dots$$
7. a) Solve $\frac{dy}{dx} = 1 - y$, $y(0) = 0$ using Euler's method. Find y at $x=0.1$ and $x=0.2$. Compare the result with exact result. 6
- b) Test the given series whether it converges or not. $\frac{0.1}{2}$ 4+4
 - i) $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n+1} + \sqrt{n}}$
 - ii) $\sum_{n=1}^{\infty} \left(\frac{n}{n+1}\right)^{n^2}$